

Лабораторная работа № 3

Измерение и тестирование пропускной способности сети. Воспроизводимый эксперимент

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Основной целью работы является знакомство с инструментом для измерения пропускной способности сети в режиме реального времени — iPerf3, а также получение навыков проведения воспроизводимого эксперимента по измерению пропускной способности моделируемой сети в среде Mininet.

```
mininet@mininet-vm:~/work/lab_iperf3$ cd ~/work/lab_iperf3
mininet@mininet-vm:~/work/lab_iperf3$ mkdir lab_iperf3_topo
mininet@mininet-vm:~/work/lab_iperf3$ cd ~/work/lab_iperf3/lab_iperf3_topo
mininet@mininet-vm:~/work/lab_iperf3/lab_iperf3_topo$ cp ~/mininet/examples/emphynet.py ~/work/lab_iperf3/lab_iperf3_topo
mininet@mininet-vm:~/work/lab_iperf3/lab_iperf3_topo$ mv emphynet.py lab_iperf3_topo.py
mininet@mininet-vm:~/work/lab_iperf3/lab_iperf3_topo$
```

Рис. 1: Копирование файла emphynet.py

Выполнение лабораторной работы

```
mininet@mininet-vm: ~/work/lab_iperf3/lab_iperf3_topo
GNU nano 4.8 lab_iperf3_topo.py
#!/usr/bin/env python

"""
This example shows how to create an empty Mininet object
(without a topology object) and add nodes to it manually.
"""

from mininet.net import Mininet
from mininet.node import Controller
from mininet.cli import CLI
from mininet.log import setLogLevel, info

def emptyNet():

    "Create an empty network and add nodes to it."

    net = Mininet( controller=Controller, waitConnected=True )

    info( '*** Adding controller\n' )
    net.addController( 'c0' )

    info( '*** Adding hosts\n' )
    h1 = net.addHost( 'h1', ip='10.0.0.1' )
    h2 = net.addHost( 'h2', ip='10.0.0.2' )

    info( '*** Adding switch\n' )
    s3 = net.addSwitch( 's3' )

    info( '*** Creating links\n' )
    net.addLink( h1, s3 )
    net.addLink( h2, s3 )
```

```
*** Adding controller
*** Adding hosts
*** Adding switch
*** Creating links
*** Starting network
*** Configuring hosts
h1 h2
*** Starting controller
c0
*** Starting 1 switches
s3 ...
*** Waiting for switches to connect
s3
*** Running CLI
*** Starting CLI:
mininet> net
h1 h1-eth0:s3-eth1
h2 h2-eth0:s3-eth2
s3 lo: s3-eth1:h1-eth0 s3-eth2:h2-eth0
c0
mininet> inks
*** Unknown command: inks
mininet> links
h1-eth0<->s3-eth1 (OK OK)
h2-eth0<->s3-eth2 (OK OK)
mininet> dump
<Host h1: h1-eth0:10.0.0.1 pid=778>
<Host h2: h2-eth0:10.0.0.2 pid=782>
<OVSSwitch s3: lo:127.0.0.1,s3-eth1:None,s3-eth2:None pid=787>
<Controller c0: 127.0.0.1:6653 pid=771>
```

Выполнение лабораторной работы

```
mininet@mininet-vm: ~/work/lab_iperf3/lab_iperf3_topo
GNU nano 4.8 lab_iperf3_topo.py
#!/usr/bin/env python

"""
This example shows how to create an empty Mininet object
(without a topology object) and add nodes to it manually.
"""

from mininet.net import Mininet
from mininet.node import Controller
from mininet.cli import CLI
from mininet.log import setLogLevel, info

def emptyNet():

    "Create an empty network and add nodes to it."

    net = Mininet( controller=Controller, waitConnected=True )

    info( '*** Adding controller\n' )
    net.addController( 'c0' )

    info( '*** Adding hosts\n' )
    h1 = net.addHost( 'h1', ip='10.0.0.1' )
    h2 = net.addHost( 'h2', ip='10.0.0.2' )

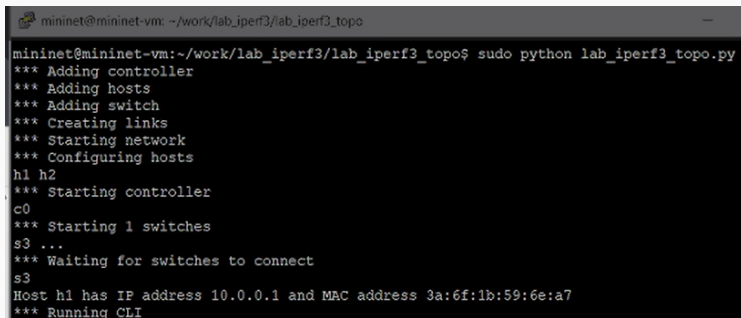
    info( '*** Adding switch\n' )
    s3 = net.addSwitch( 's3' )

    info( '*** Creating links\n' )
    net.addLink( h1, s3 )
    net.addLink( h2, s3 )

    info( '*** Starting network\n' )
    net.start()

    print( "Host", h1.name, "has IP address", h1.IP(), "and MAC address", h1.MAC() )
    print( "Host", h2.name, "has IP address", h2.IP(), "and MAC address", h2.MAC() )
    info( '*** Running CLI\n' )
    CLI( net )

    info( '*** Stopping network' )
    net.stop()
```



```
mininet@mininet-vm: ~/work/lab_iperf3/lab_iperf3_topo
mininet@mininet-vm:~/work/lab_iperf3/lab_iperf3_topo$ sudo python lab_iperf3_topo.py
*** Adding controller
*** Adding hosts
*** Adding switch
*** Creating links
*** Starting network
*** Configuring hosts
h1 h2
*** Starting controller
c0
*** Starting 1 switches
s3 ...
*** Waiting for switches to connect
s3
Host h1 has IP address 10.0.0.1 and MAC address 3a:6f:1b:59:6e:a7
*** Running CLI
```

Рис. 5: Проверка работы внесенных изменений

Выполнение лабораторной работы

```
"""
This example shows how to create an empty Mininet object
(without a topology object) and add nodes to it manually.
"""

from mininet.node import CPULimitedHost
from mininet.link import TCLink
from mininet.net import Mininet
from mininet.node import Controller
from mininet.cli import CLI
from mininet.log import setLogLevel, info

def emptyNet():

    "Create an empty network and add nodes to it."

    net = Mininet( controller=Controller, waitConnected=True,
                   waitConnected=True, host = CPULimitedHost, link = TCLink )

    info( '*** Adding controller\n' )
    net.addController( 'c0' )

    info( '*** Adding hosts\n' )
    h1 = net.addHost( 'h1', ip='10.0.0.1', cpu=50 )
    h2 = net.addHost( 'h2', ip='10.0.0.2', cpu=45 )

    info( '*** Adding switch\n' )
    s3 = net.addSwitch( 's3' )

    info( '*** Creating links\n' )
    net.addLink( h1, s3, bw=10, delay='5ms', max_queue_size=1000, loss=10, use_htb=True )
    net.addLink( h2, s3 )

    info( '*** Starting network\n' )
    net.start()

    print( "Host 1", h1.name, "has IP address", h1.IP(), "and MAC address", h1.MAC() )
    print( "Host 2", h1.name, "has IP address", h2.IP(), "and MAC address", h2.MAC() )
```

Выполнение лабораторной работы

```
*** Adding controller
*** Adding hosts
*** Adding switch
*** Creating links
*** Starting network
*** Configuring hosts
h1 h2
*** Starting controller
c0
*** Starting 1 switches
s3 ...
*** Waiting for switches to connect
s3
Host 1 h1 has IP address 10.0.0.1 and MAC address 7e:bb:71:20:6a:8d
Host 2 h1 has IP address 10.0.0.2 and MAC address 42:8c:e6:3d:f2:98
*** Running CLI
*** Starting CLI:
mininet> links
h1-eth0<->s3-eth1 (OK OK)
h2-eth0<->s3-eth2 (OK OK)
mininet> dump
<Host h1: h1-eth0:10.0.0.1 pid=1006>
<Host h2: h2-eth0:10.0.0.2 pid=1008>
<OVSSwitch s3: lo:127.0.0.1,s3-eth1:None,s3-eth2:None pid=1013>
<Controller c0: 127.0.0.1:6653 pid=999>
mininet> exit
*** Stopping network*** Stopping 1 controllers
c0
*** Stopping 2 links
..
*** Stopping 1 switches
s3
*** Stopping 2 hosts
h1 h2
*** Done
mininet@mininet-vm:~/work/lab_iperf3/lab_iperf3_topo$ cp lab_iperf3_topo.py lab_iperf3_topo2.py
```

```
mininet@mininet-vm:~/work/lab_iperf3/lab_iperf3_topo$ cp lab_iperf3_topo.py lab_iperf3_topo2.py
mininet@mininet-vm:~/work/lab_iperf3/lab_iperf3_topo$ nano lab_iperf3_topo2.py
mininet@mininet-vm:~/work/lab_iperf3/lab_iperf3_topo$ sudo python lab_iperf3_topo2.py
File "lab_iperf3_topo2.py", line 21
```

Рис. 8: Создание копии скрипта lab_iperf3_topo2.py

Выполнение лабораторной работы

```
import time
from mininet.node import CPULimitedHost
from mininet.link import TCLink
from mininet.net import Mininet
from mininet.node import Controller
from mininet.cli import CLI
from mininet.log import setLogLevel, info

def emptyNet():

    "Create an empty network and add nodes to it."

    net = Mininet( controller=Controller, waitConnected=True, host = CPULimitedHost, link =

    info( '*** Adding controller\n' )
    net.addController( 'c0' )

    info( '*** Adding hosts\n' )
    h1 = net.addHost( 'h1', ip='10.0.0.1' )
    h2 = net.addHost( 'h2', ip='10.0.0.2' )

    info( '*** Adding switch\n' )
    s3 = net.addSwitch( 's3' )

    info( '*** Creating links\n' )
    net.addLink( h1, s3, bw=100, delay='75ms' )
    net.addLink( h2, s3, bw=100, delay='75ms' )

    info( '*** Starting network\n' )
    net.start()
    info( '*** Starting network\n' )

    info( '*** Traffic generation\n' )
    h2.cmdPrint( 'iperf3 -s -D -1' )
    time.sleep(10) # Wait 10 seconds for servers to start
    h1.cmdPrint( 'iperf3 -c', h2.IP(), '-J >iperf_result.json' )

    # info( '*** Running CLI\n' )
    # CLI( net )
```

```
mininet@mininet-vm:~/work/lab_iperf3/iperf3$ sudo python lab_iperf3.py
*** Adding controller
*** Adding hosts
*** Adding switch
*** Creating links
(100.00Mbit 75ms delay) (100.00Mbit 75ms delay) (100.00Mbit 75ms delay) (100.00Mbit 75ms delay)
*** Starting network
*** Configuring hosts
h1 (cfs -1/100000us) h2 (cfs -1/100000us)
*** Starting controller
c0
*** Starting 1 switches
s3 (100.00Mbit 75ms delay) (100.00Mbit 75ms delay) ... (100.00Mbit 75ms delay) (100.00Mbit 75ms delay)
*** Waiting for switches to connect
s3
*** Starting network
*** Traffic generation
*** h2 : ('iperf3 -s -D -1',)
*** h1 : ('iperf3 -c', '10.0.0.2', '-J >iperf_result.json')
*** Stopping network*** Stopping 1 controllers
c0
*** Stopping 2 links
..
*** Stopping 1 switches
s3
*** Stopping 2 hosts
h1 h2
*** Done
```

Рис. 10: Запуск скрипта lab_iperf3.py

```
GNU nano 4.8

all: iperf_result.json plot

iperf_result.json:
    sudo python lab_iperf3.py

plot: iperf_result.json
    I plot_iperf.sh iperf_result.json

clean:
    -rm -f *.json *.csv
    -rm -rf results
```

Рис. 11: Создание Makefile

В результате выполнения данной лабораторной работы я познакомилась с инструментом для измерения пропускной способности сети в режиме реального времени — iPerf3, а также получила навыки проведения воспроизводимого эксперимента по измерению пропускной способности моделируемой сети в среде Mininet.